

POWERBOX · DESIGN PRECIS

PowerBox design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ 0a81f90		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.4	2026-09-25	Amish Chadha	TRL 3. SwapCell interface v0.3 (wake, station mode, latch class); Amish's decisions recorded (PBX-DDR-001); numbers checked against PBX-CAL-001; 30 A buck; general arrangement PBX-DWG-001
0.3	2026-09-24	Amish Chadha	Removed StepGen as a charging source after StepGen became a walking vehicle (Amish, 2026-09-24)
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, input and output specification, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

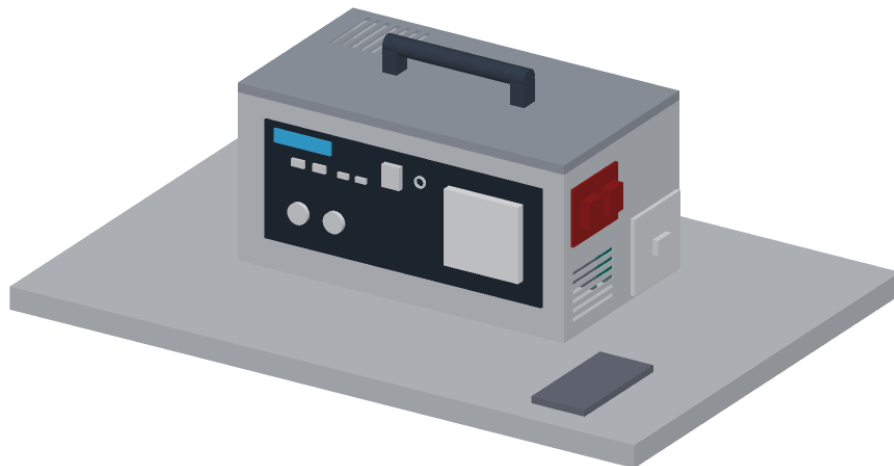
PowerBox design precis

PowerBox is an aluminium carry case about the size of a small toolbox (460 x 260 x 220 mm body, 482 x 276 x 278 mm overall) with a SwapCell pack sliding into a bay through an end door. A charge controller takes DC from a solar panel, a certified external charger takes grid power, and a charged pack from a SunSpoke bike can simply be swapped in. Power leaves through USB-C PD, USB-A, 12 V sockets and one 300 W, 230 V pure sine AC outlet behind a 30 mA RCD. It never connects to household wiring. The sizing note PBX-CAL-001 shows one pack gives 419 Wh usable, which runs the reference evening of lights, phones, radio and a router **1.87 times, short of the two evenings in R2**. The box weighs about 8.6 kg with the pack, and the PowerBox parts cost about \$449 without the pack, against a \$450 budget.

PowerBox builds to **SwapCell interface v0.3**. It is a station host (item C), so it can run loads while charging from solar; its receptacle carries the 10 k Ω INTERLOCK coding resistor that wakes a sleeping pack (item W); and its bay uses a class D latch catch (item V).

PowerBox

CONCEPT, NOT FOR FABRICATION



Grey: Table top, phone for scale

Figure 1. PowerBox on a table, with a phone for scale. Output panel on the front, input panel and pack bay door on the right end. Rendered from the parametric model `cad/src/model.py`.

■ How it works

1. **Charge.** Energy comes in through one of three paths, all controlled by the host controller: - **DC input** (a solar panel or another 12 to 60 V DC source, up to 200 W) goes through a buck-boost charge controller that tracks the source's maximum power point and charges the pack at up to 54.6 V. - **Grid** goes through a certified external 54.6 V, 5 A charger brick with a DC plug. PowerBox itself has no AC inlet, so it cannot be plugged into anything that would back-feed. - **SunSpoke** charges the pack while riding or at its own 100 W panel. The rider brings the charged SwapCell pack home and swaps it in. SunSpoke's panel can also plug into the DC input directly. StepGen, a walking vehicle, runs on the same SwapCell pack, so a pack can move between PowerBox and StepGen in the same way.
2. **Store.** The SwapCell pack (13S2P lithium-ion, 46.8 V nominal, 468 Wh nameplate) lies on one of its guide faces on floor runners, with its latch face toward the front of the case, and mates through the SwapCell blind-mate connector at the far end of the bay. A class D catch on a bracket at the front of the bay takes the pack's latch pawl, and the closed door stops the handle.
3. **Wake and handshake.** Seating the pack closes the INTERLOCK loop through the receptacle's 10 kΩ coding resistor, which wakes the pack. With no heartbeat after 2 s the pack enters legacy discharge (15 A limit), which powers the host; the host then sends a station heartbeat (host type 3) requesting mode 2 (discharge) or mode 4 (charge-discharge). A pack that fell asleep in the bay is woken with a recessed wake button on the front panel, which briefly opens the INTERLOCK loop.
4. **Convert and deliver.** The 39 to 54.6 V bus feeds a 48 V to 12 V, 30 A buck converter for the 12 V sockets and the USB-C PD and USB-A modules, and a 48 V input, 300 W pure sine inverter for the AC outlet. The inverter is switched in through its own relay and pre-charge resistor.
5. **Charge while in use.** When a source and a load are both present the host requests mode 4 and keeps net charge current at or below 5.0 A. With the grid charger connected (a fixed 5 A), the host limits the solar controller to the present load current.
6. **Inform.** A small display shows state of charge, input and output power and estimated time remaining, read from the pack's CAN messages. The host switches the inverter off after 10 min below 5 W and returns to off after 30 min idle.
7. **Protect.** A 30 A main fuse, a 20 A breaker on the inverter feed, per-output fuses, thermostatic fan control and the pack's own BMS limits guard against faults.

PowerBox: energy per usable cycle, DC input to loads (estimates, PBX-CAL-001)

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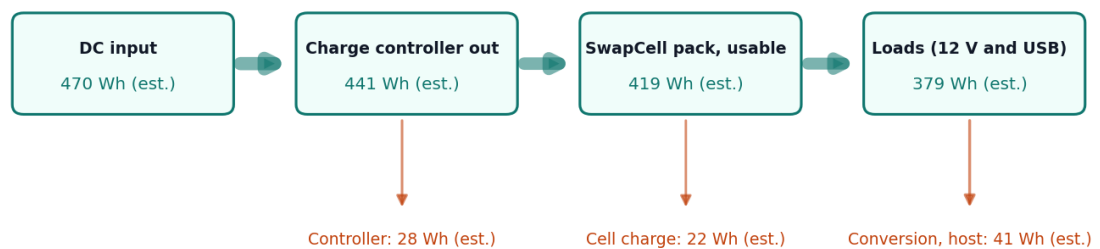


Figure 2. Energy per usable cycle, DC input to loads, from PBX-CAL-001. All values are estimates: controller 94 %, cell charge 95 %, output conversion for the reference evening mix of 12 V and USB loads.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Choice	Notes
1	Enclosure body	Folded 1.2 mm aluminium tub, 460 x 260 x 220 mm	Spreads heat and resists fire better than plastic. Decided by Amish, 2026-09-25
2	Lid	1.2 mm aluminium with a 12 mm skirt and exhaust slots at the back	Removable for service
3	Carry handle	Folding bar handle on the lid, 200 mm span	Rated 15 kg or more
4	SwapCell pack	SwapCell interface v0.3 reference pack, 468 Wh, 2.85 kg	Priced once in the SwapCell BOM, not here. Removable pack decided by Amish, 2026-09-25
5	Pack bay	Floor runners and top guide rail (1 mm clearance per side), class D catch bracket, shelf above, SwapCell receptacle on a floating mount with a 10 kΩ INTERLOCK coding resistor	Horizontal side-loading bay decided by Amish, 2026-09-25
6	Pack bay door	Hinged door on the right end, magnetic catch, padlock eye	Second stop behind the pack handle
7	Multi-input charge controller	Buck-boost DC-DC, 12 to 60 V in, 200 W, CC-CV output set by the host	One DC input decided by Amish, 2026-09-25; MPPT in host firmware
8	Host controller	ESP32 with CAN transceiver, relays and current sensors	SwapCell station host; charge control, auto-off, display
9	Inverter	300 W pure sine, 48 V input, 600 W for 1 s, 230 V 50 Hz, output isolated so the neutral can be bonded	AC region decided by Amish, 2026-09-25
10	DC-DC converter	48 V to 12 V, 30 A buck	Resized from 20 A by PBX-CAL-001
11	Output panel	USB-C PD 100 W and 60 W, 2 x USB-A, 2 x 12 V sockets, main switch, recessed wake button	Front face
12	AC outlet	Single 230 V outlet behind a 30 mA RCD	Only mains-voltage point on the box
13	Display	2.4 in TFT or e-paper	State of charge, power in and out, time left
14	Input panel	2 x Anderson Powerpole PP45: DC in, charger in	Right end, above the bay door
15	Exhaust fan	80 mm 12 V fan, thermostatic; filtered intake slots on the right end	Pulls air across the case and out of the left end and lid

Items 16 (protection and wiring), 17 (grid charger brick) and 18 (hardware) are in the BOM but not modelled. The general arrangement is drawing PBX-DWG-001 (`cad/drawings/PBX-DWG-001.pdf`), generated from `cad/src/model.py`.

PowerBox: cutaway

CONCEPT, NOT FOR FABRICATION

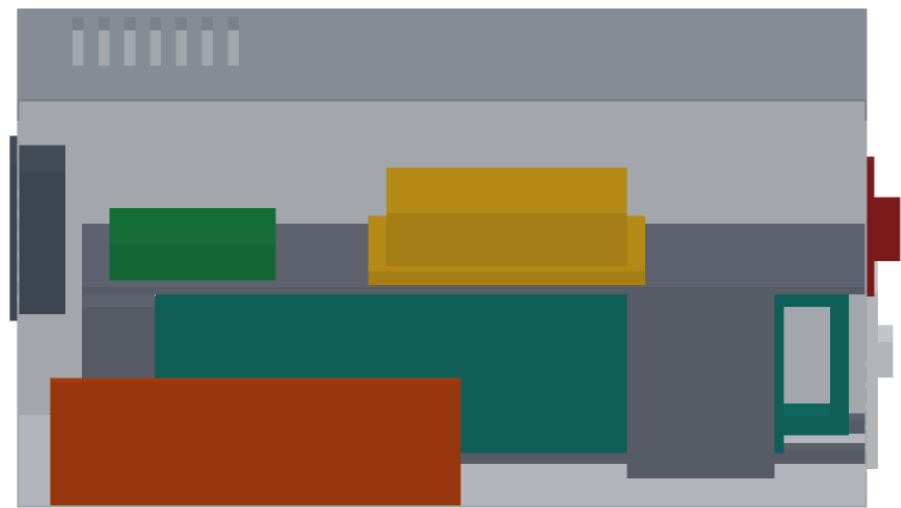


Figure 3. Section looking from the front: inverter (9) on the floor at the front, SwapCell pack (4) in its bay at the back, charge controller (7) and host (8) on the shelf above, fan (15) on the left end, input panel (14) and bay door (6) on the right end.

■ **Key numbers**

All values come from PBX-CAL-001 (docs/04-calcs/sizing.py) and are paper estimates. Assumptions: the SwapCell v0.3 reference pack (466 Wh at 0.2C, 39.0 to 54.6 V); 10 to 100 % state of charge; 12 V buck 94 %, USB modules 93 % from 12 V, charge controller 94 % nominal, cell charge 95 %, grid charger 90 %; inverter efficiency and idle draw typical of 48 V, 300 W pure sine units (no datasheet yet).

Table 2. Energy, charge times, size and cost.

Quantity	Value	Basis	Requirement
Usable energy	419 Wh (407 Wh with minimum cells)	90 % of 466 Wh	R1 met (400 Wh)
Reference evening (Table 1 of PBX-REQ-001)	225 Wh from the pack	165 Wh for 12 V loads at 94 %, 55 Wh for USB at 87 %, 5 Wh host and idle converters over 5 h	
Evenings per pack	1.87	419 / 225	R2 not met (target 2)
Grid charge, 10 to 100 %	2.1 h; 492 Wh from the wall	1.5 h CC at 5 A, then 0.6 h CV; charger 90 %	R4 met (3 h)
Solar, 200 W panel	634 Wh per clear day at the pack; full in 0.70 day	4.5 peak sun hours, 0.75 derating, 94 % controller	R5 met
Solar, SunSpoke 100 W panel	317 Wh per clear day; one evening in 0.75 day	Same basis	
Charge controller efficiency	90.5 % at 12 V and 200 W; 93 % or more elsewhere from 40 to 200 W	Loss model, no datasheet	R6 at risk
Station mode, net charge	4.48 A worst case; 2.74 A typical with the router	200 W at 94 % into 42 V; limit 5.0 A	Within the SwapCell limit
12 V bus demand, all DC outputs	318 W, 26.5 A	120 W sockets plus 184 W USB at 93 %	30 A buck (was 20 A)
Peak pack current	11.5 A at 400 W output; 20.9 A during a 1 s inverter surge	At 39.0 V	Within 20 A continuous and 35 A peak
Inverter idle, if left on	192 Wh per day, 46 % of usable	8 W for 24 h	Auto-off after 10 min (R8)
Standby, off	3.2 % per month	2.5 % self-discharge plus 100 µA BMS sleep	R8 met (5 %)
Standby, ready	0.6 W, 14.4 Wh per day	ESP32 0.2 W, CAN and BMS 0.2 W, display 0.2 W	R8 met (1.0 W); host returns to off after 30 min idle
Heat at full load while charging	60 W; air rise about 6 K with the fan	Inverter 41 W, buck 6.4 W, controller 12 W, host 1 W	Fan needed
Mass	8.6 kg (19.0 lb) with pack	Pack 2.85, body 1.41, lid 0.44, inverter 1.20, other parts 2.71 kg	R10 met (10 kg); charger brick about 0.8 kg extra
Size	482 x 276 x 278 mm overall	Parametric model	R10 met, 2 mm spare in height
PowerBox parts cost	\$449	bom/bom.csv , indicative prices	R11 at risk (\$1 margin)
Cost with one SwapCell pack	\$863, for reference	\$449 plus \$414 from the SwapCell BOM	Pack excluded from this budget (decided)

Table 3. Runtime on one full pack (419 Wh usable) for single loads.

Load	Output	Power at the load	Drawn from the pack	Runtime
Wi-Fi router and fibre terminal	12 V DC	12 W	12.8 W	about 33 h
Three 5 W LED bulbs	12 V DC	15 W	16.0 W	about 26 h
Radio	12 V DC	5 W	5.3 W	about 79 h
Phone charges	USB	12 Wh each	13.7 Wh each	about 31 charges
Laptop	USB-C PD	45 W	51.5 W	about 8 h
Small LED TV	AC	60 W	70.6 W	about 6 h
Desk fan	AC	25 W	32.1 W	about 13 h
Maximum AC load	AC	300 W	340.9 W	about 1.2 h
Refrigerator or kettle	AC	Out of scope: starting surge or power beyond the 300 W inverter		

Table 4. Input and output specification.

Port	Type and connector	Voltage	Power or current	Notes
DC in	Anderson PP45, red and black	12 to 60 V DC	200 W maximum, 16.7 A at 12 V	Solar panel or SunSpoke panel; MPPT; TVS and reverse-polarity protection; 20 A fuse
Charger in	Anderson PP45, keyed differently from DC in	54.6 V DC from the certified brick	5 A	Grid charging only through the external charger
Pack	SwapCell blind-mate connector, interface v0.3	39.0 to 54.6 V	20 A continuous available; PowerBox uses 11.5 A or less	CAN 2.0B at 250 kbit/s; 120 Ω termination in PowerBox; 10 k Ω INTERLOCK coding resistor; station host type 3
USB-C PD 1	USB-C	5 to 20 V	100 W	Laptops
USB-C PD 2	USB-C	5 to 20 V	60 W	Phones, tablets
USB-A 1, 2	USB-A	5 V	12 W each	Phones, lamps, radios
12 V DC 1	Car socket	12 V regulated	10 A shared with 12 V DC 2	Router, lights, radio
12 V DC 2	5.5 x 2.1 mm barrel	12 V regulated	As above	
AC out	One 230 V socket behind a 30 mA RCD	230 V 50 Hz, pure sine	300 W continuous, 600 W for 1 s	Auto-off; never to be connected to household wiring
Total output			400 W	Host limit to keep pack current and heat in range

■ Key design choices

All of these were decided by Amish on 2026-09-25 (go with recommendation), PBX-DDR-001, unless marked otherwise.

- **Removable SwapCell pack, with a fixed internal pack as fallback.** One battery moves between PowerBox, SunSpoke and the SwapCell dock, and a worn pack is replaced without discarding the box. The fixed 12.8 V LiFePO4 pack (about 30 Ah, 384 Wh, about \$110) with a 12 V inverter and no CAN stays only as a documented fallback.
- **Side-loading horizontal bay.** Laying the pack along the case length with a door on the end keeps the case low and stable; a top-loading bay would make it about 450 mm tall.
- **One DC input with maximum power point tracking.** A second independent input (about \$30 more) only if users need two panels at once.
- **Grid charging only through a certified external charger.** No mains-input electronics in the box and no AC inlet. The brick can be omitted where a SwapCell dock is on hand (\$399 of parts).
- **A 300 W, 230 V AC outlet, with a DC-only variant.** The DC-only variant saves about \$92 and removes all mains voltage from the box. 120 V 60 Hz is a later variant.
- **Folded aluminium enclosure.**

- **No human-powered input now.** A pedal generator could later be a separate repo that plugs into the DC input.
- **Station host on SwapCell interface v0.3.** Wake through the coded INTERLOCK loop (item W), charge-discharge mode 4 as a station host (item C) and a class D latch catch with the door as a second stop (item V). These replace the keep-alive cell and local pass-through workarounds considered at TRL 2.
- **Budget.** `budget_usd` stays at \$450 and covers the PowerBox parts, including the grid charger; the SwapCell pack is priced once in the SwapCell BOM.
- **Wake button, 30 A buck and fuse ratings.** Proposed in PBX-CAL-001 and decided by Amish, 2026-09-25: go with recommendation (PBX-DDR-002). A recessed, normally closed wake button in series with the 10 k Ω coding resistor; a 30 A (360 W) 48 V to 12 V buck; every 48 V fuse rated 60 V DC or more with at least 1 kA breaking capacity, as in PBX-CAL-001 Table 3.

PowerBox: exploded view

CONCEPT, NOT FOR FABRICATION

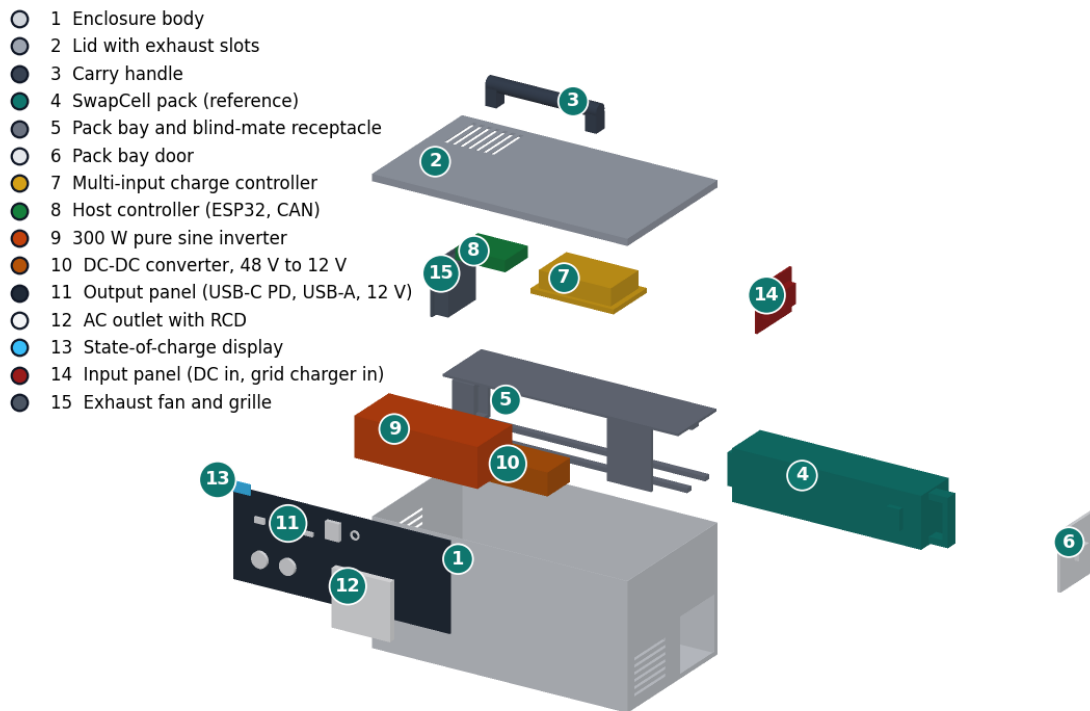


Figure 4. Exploded view with BOM numbers.

■ Safety

Safety: PowerBox is standalone only. Never connect the AC outlet to a wall outlet, household wiring, a distribution board or a transfer switch. Back-feeding can kill utility workers repairing lines during an outage, can damage the box and the home's wiring, and is illegal.

Safety: The SwapCell pack is a 468 Wh lithium-ion battery that can deliver about 500 A into a short. Use it only with the SwapCell BMS, keep the pack's fuse and PowerBox's 30 A main fuse in place, and never charge a damaged, swollen or wet pack. Charge on a non-combustible surface, away from exits, beds and children, and do not cover the box while charging. Keep a smoke alarm in the room.

Safety: The AC outlet carries 230 V. It is behind a 30 mA RCD, and for the RCD to work the inverter's output neutral must be bonded to the case and the protective earth pin, as in a vehicle or boat installation. This needs an inverter whose output is isolated from its DC input; many small inverters float or are not isolated. This must be confirmed from the chosen inverter's datasheet before any build.

- **No back-feed by design.** There is no AC inlet on the box, and grid charging uses a certified external charger with a DC plug, so no cable can join PowerBox's AC output to a live circuit through an ordinary plug. The user guide and labels must still warn against double-male cords and improvised connections.
- **Extra-low-voltage bus.** The battery side stays at 54.6 V or less, below the 60 V DC limit for extra-low voltage, so only the inverter output and the charger brick input are at mains voltage.
- **Fuse ratings.** Every fuse on the 48 V side must be rated for 60 V DC or more and at least 1 kA breaking capacity, because a short at the pack could draw about 496 A (PBX-CAL-001 section 6).
- **Coded interlock.** The receptacle must carry the 10 k Ω coding resistor. A direct link from INTERLOCK to SGND reads as a fault and the pack stays dead, which fails safe. The wake button opens the loop, so pressing it while the box is running cuts the output briefly; it is recessed to prevent accidental presses.
- **Charging while in use.** In mode 4 PowerBox acts as a charger. The host keeps net charge at or below 5.0 A and the pack voltage at or below 54.6 V, and limits the solar controller whenever the fixed 5 A grid charger is connected. The pack refuses charge below 0 °C and above 45 °C cell temperature.
- **Ventilation.** At full load while charging the case holds about 60 W of heat. A thermostatic fan pulls air through filtered intake slots on the right end and out through the left end and lid. The host derates or shuts outputs down above an internal temperature limit, and the pack enforces its own limits. The case must not be used in a closed cupboard or bag.
- **Cords and connectors.** Keyed Powerpole connectors prevent the charger and DC inputs from being swapped. Input and output cords must be rated for their current, kept out of walkways and never run under rugs. The 12 V sockets are fused at 10 A.
- **Pack handling.** The bay door keeps fingers away from the connector, and the SwapCell interlock keeps the pack output dead until it is fully seated.
- **Not a medical or life-safety supply.** PowerBox must not be relied on for oxygen concentrators or other life-support equipment.

■ Open questions

Items that remain open after PBX-DDR-001 and PBX-DDR-002. None of them is TRL 4 work to be started now; TRL 4 is on hold by Amish's instruction.

- **R2 shortfall (1.87 evenings).** Relax R2, discharge to 3.5 % (a 5 % cut-off gives only 1.97 evenings), or revise the evening profile with users (about 189 Wh at the loads). Proposed, awaiting Amish.
- **Legacy-to-station transition.** SwapCell interface v0.3 does not say whether a pack in legacy discharge accepts a station heartbeat and moves to mode 2 or 4 without opening its output. PowerBox's host is powered from the pack and depends on this. Raised with SwapCell; not changed locally.

- **Inverter selection.** Confirm idle draw, light-load efficiency and an isolated output whose neutral can be bonded for the RCD.
- **Charge controller selection (R6, at risk).** Confirm 90 % or more at 12 V and 200 W and at 40 W, and stable tracking.
- **Budget margin (R11, at risk).** \$1 of margin on indicative prices; named supplier quotes are needed.
- **Theft resistance and energy metering for charging points.** Proposed, awaiting Amish.
- **First co-design partner.** Left open; the portfolio picks partners per area later. The load profile, pack-swap routine and price must be validated with users.

Concept media: [blueprint sheet](#), [interactive 3D model](#).